**Name : Deepa Sunil Pandey** **Batch : T21**
**Roll No : 79**

# EXPERIMENT 5

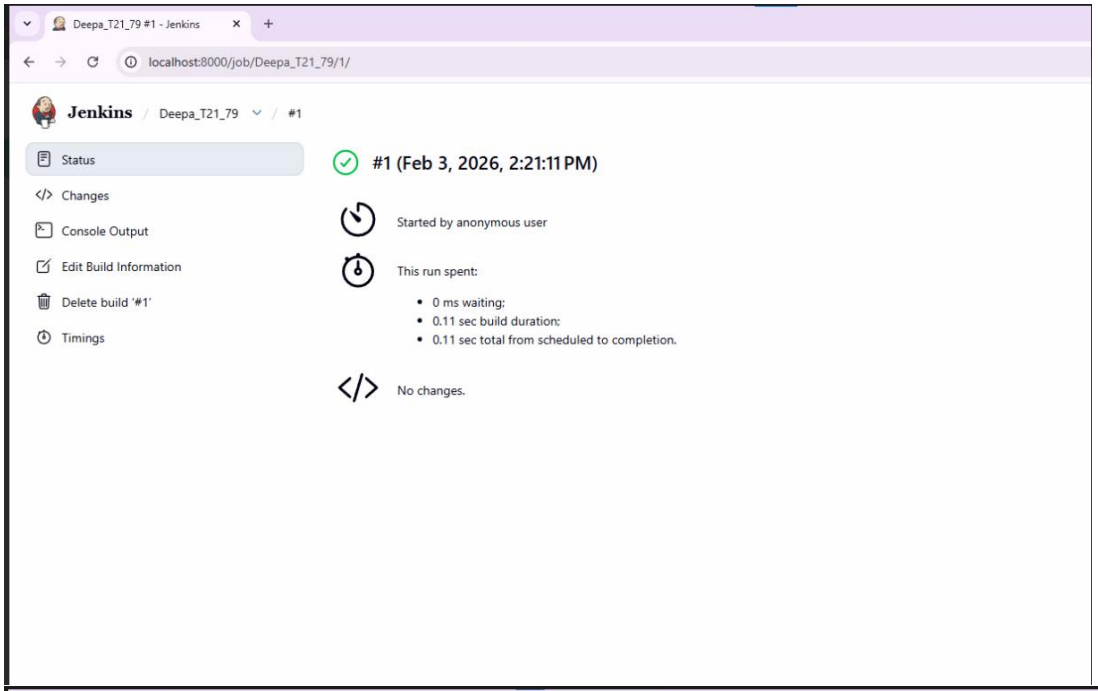
**Aim :** To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

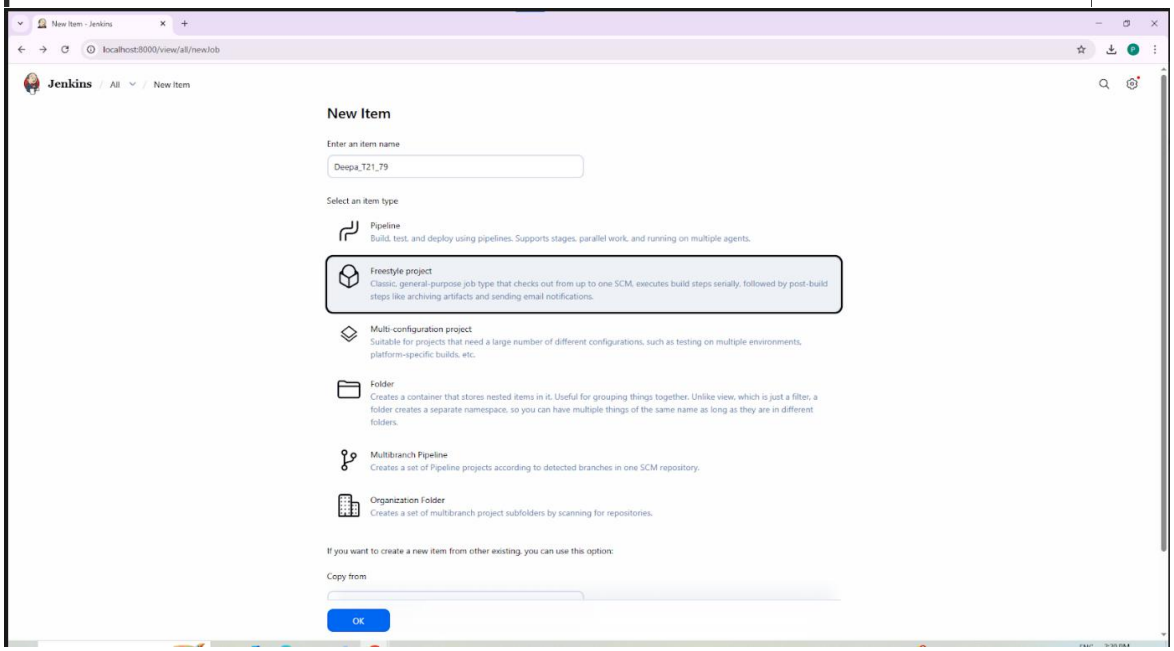
## Theory :

Continuous Integration and Continuous Deployment (CI/CD) is a software development practice that automates the process of building, testing, and deploying applications. Jenkins is a widely used open-source automation server that enables developers to create pipelines for integrating code changes, running tests, and deploying applications efficiently. In this experiment, build tools such as Apache Maven, Gradle, or Apache Ant are used to compile source code, manage dependencies, and package the application.

A Jenkins pipeline is defined using a script (commonly called a Jenkinsfile), which specifies stages such as Build, Test, and Deploy. During the build stage, the selected tool (Maven/Gradle/Ant) compiles the project and generates an artifact (e.g., WAR file for web applications). In the testing stage, automated test cases are executed to ensure code quality. Finally, in the deployment stage, the generated artifact is deployed to a web server such as Apache Tomcat, which hosts and runs Java-based web applications.

This pipeline-based approach ensures faster development cycles, early detection of errors, and consistent deployment practices. By integrating build tools with Jenkins and deploying on Tomcat, organizations achieve automation, scalability, and reliability in their software delivery process.


Deepa_T21_79 #1 - Jenkins
localhost:8000/job/Deepa_T21_79/1/
Jenkins / Deepa_T21_79 / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
#1 (Feb 3, 2026, 2:21:11 PM)
Started by anonymous user
This run spent:
0 ms waiting;
0.11 sec build duration;
0.11 sec total from scheduled to completion.
No changes.


New Item - Jenkins
localhost:8000/view/all/newJob
Jenkins / All / New Item
New Item
Enter an item name
Deepa_T21_79
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
If you want to create a new item from other existing, you can use this option:
Copy from
OK

### 3 Add Build Step

- Scroll to **Build**
- Click **Add build step**
- Select **Execute Windows batch command**

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### 4 Write command

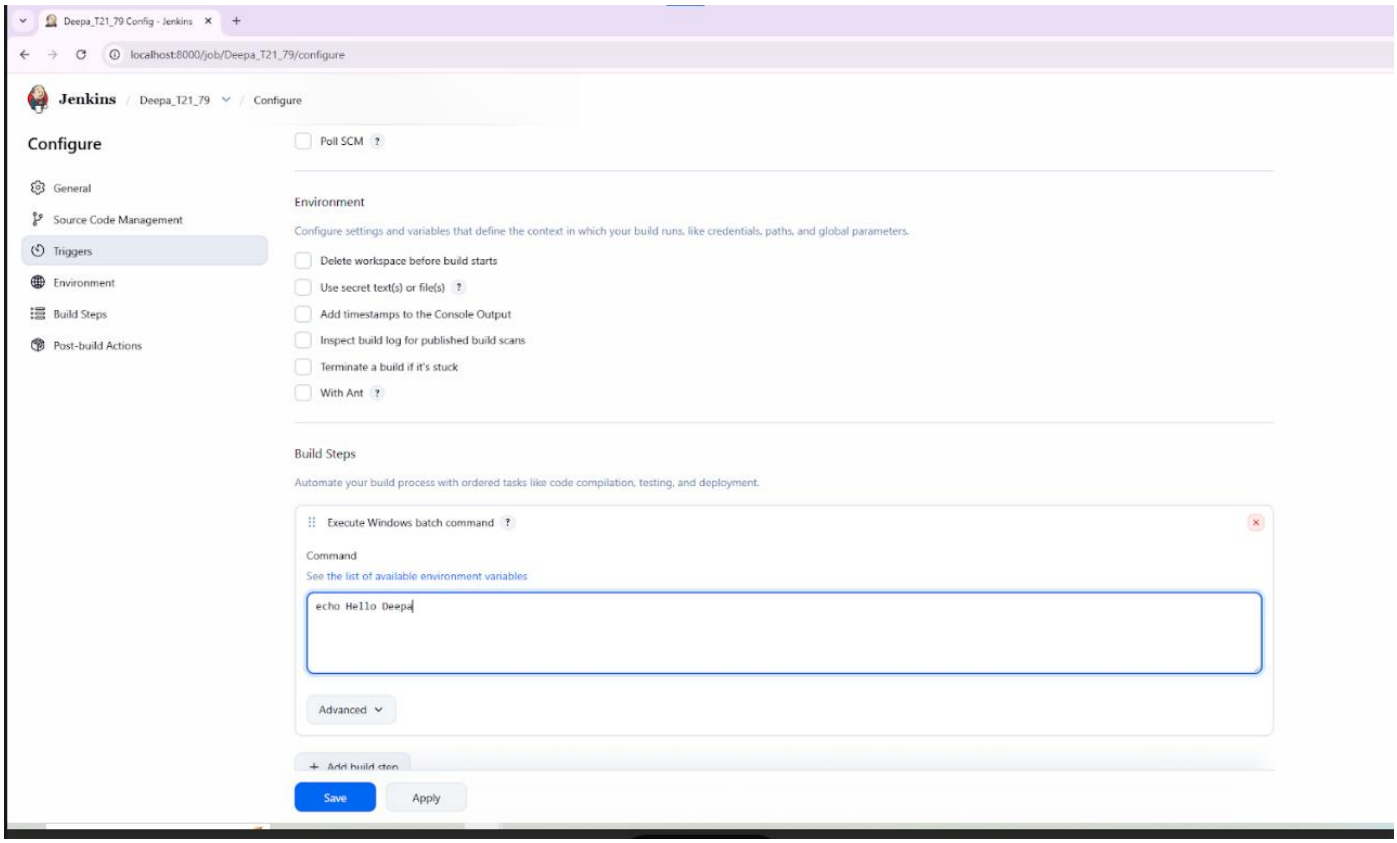
```bat
echo Hello Jenkins
```

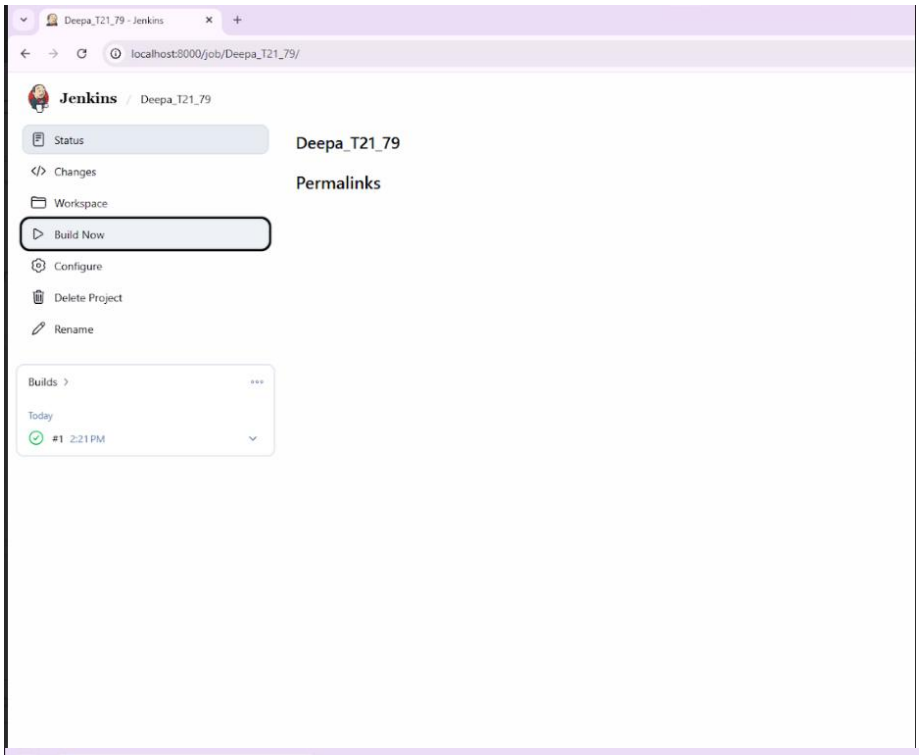
- Click **Apply → Save**

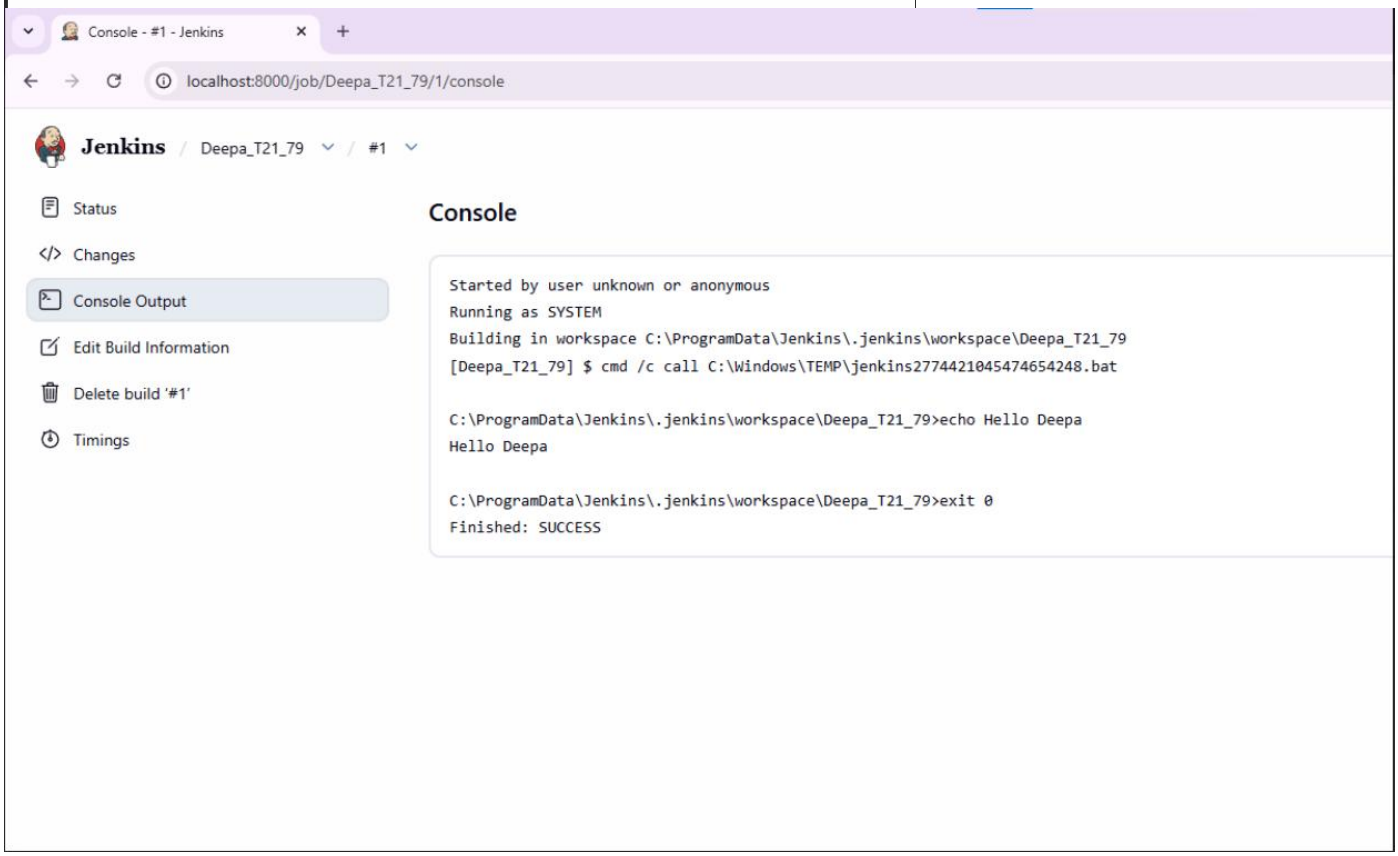
---

### 5 Build and Output

- Click **Build Now**






**Coclusion :** Successfully implemented the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server